**Tutorial 6: Hypothesis testing**

1. A random sample of size $n = 25$ is taken from a $\mathrm{N}(\mu, 900)$ population in order to test the hypothesis $H_0 : \mu = 50$ vs $H_1 : \mu > 50$ at the 5% significance level.

   (a) Determine the critical (rejection) region of this test in terms of the mean of the sample $\bar{X}$.

   (b) Find the probability of type II error when $\mu = 56$.

2. Consider the random variable $X$ having a Poi($\lambda$) distribution. We want to test the hypothesis $H_0 : \lambda = 2$ vs $H_1 : \lambda = 3$ based on a single observation $x$, and decide to use the critical (rejection) region given by the interval $x \geq 5$.

   (a) Calculate the probabilities of type I error and type II error of this test.

   (b) Calculate the power of the test.

3. A random sample of size 19 was taken from a normal population in order to estimate the mean $\mu$ and variance $\sigma^2$ of the population. The 19 observations were:

   | | | | | | | | | | |
   |---|---|---|---|---|---|---|---|---|---|
   | 230 | 165 | 192 | 173 | 200 | 138 | 216 | 233 | 198 | 189 |
   | 172 | 207 | 196 | 207 | 204 | 124 | 188 | 252 | 192 | |

   for which $\sum x = 3676$ and $\sum x^2 = 728234$.

   (a) Calculate a 95% CI for the mean $\mu$. State any assumptions that you make.

   (b) Test $H_0 : \mu = 200$ $v$ $H_1 : \mu \neq 200$ at the 5% level of significance. State clearly the conclusion and relate your answer to the CI found in (a).

   (c) Calculate the $p$-value of the observed test statistic in (b) and give its interpretation.

   (d) Test $H_0 : \sigma^2 = 2500$ $v$ $H_1 : \sigma^2 < 2500$ at the 1% level of significance and calculate the $p$-value of the observed test statistic (for the $p$-value use linear interpolation with the values from the tables and check your answer in `R` with the `pchisq()` command).

# Exam Style Questions

A supermarket chain claims that breakfast cereal eaters are not able to tell the difference between their own (cheaper) brand of cornflakes and a leading major brand.

To test the hypothesis that people cannot tell the difference, a statistician organises a tasting experiment where an individual is asked to taste 10 randomly presented bowls of cornflakes;

five of these being the supermarket's own brand and the other five the leading brand. The individual does not know how many cornflake samples were from either brand.

Under the null hypothesis that people cannot tell the difference, the probability of correctly identifying the brand of cornflakes purely by chance is considered to be 0.5. A one-sided test is used with $H_0 : p = 0.5$ versus $H_1 : p > 0.5$.

The statistician adopts the following decision rule for testing the hypothesis:

- Reject the null hypothesis if seven or more bowls of cornflakes are correctly identified; conclude that the individual can tell the difference.
- Do not reject the null hypothesis if less than seven bowls of cornflakes are correctly identified; conclude that the individual cannot tell the difference.

(a) Determine the probability of a type I error for this test.

[3 marks]

(b) Based on past experience we can quantify the probability p that an individual recognises the brand of cornflakes correctly. Determine the probability of a type II error assuming that the alternative hypothesis is of the form

(i) $H_1 : p = 0.7$

[2 marks]

(ii) $H_1 : p = 0.8$

[2 marks]

(c) Calculate the power of the test for each of your answers in part (b) and comment on how this changes as $p$ increases.

[3 marks]